

MIE1622 Assignment 2

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1 Mathematical Formulation

Let n denote the number of assets and let $w = (w_1, \dots, w_n)$ represent the vector of portfolio weights. The vector of expected returns is denoted by μ , and the covariance matrix of asset returns is denoted by Q . The portfolio variance is given by

$$\sigma_p^2 = w^T Q w.$$

All strategies are subject to the standard portfolio constraints

$$\sum_{i=1}^n w_i = 1, \quad w_i \geq 0.$$

1.1 Equal Risk Contributions Portfolio

The equal risk contributions (ERC) portfolio aims to allocate weights such that each asset contributes equally to the total portfolio risk. The risk contribution of asset i is defined as

$$RC_i = w_i(Qw)_i.$$

The ERC portfolio can be obtained by solving an optimization problem that minimizes the differences between individual risk contributions:

$$\min_w \sum_{i=1}^n (RC_i - RC_j)^2.$$

This formulation encourages the portfolio to distribute total risk evenly across all assets.

1.2 Leveraged Maximum Sharpe Ratio Portfolio

The maximum Sharpe ratio portfolio maximizes the ratio between excess expected return and portfolio volatility:

$$\max_w \frac{w^T \mu - r_f}{\sqrt{w^T Q w}}$$

where r_f is the risk-free rate.

In the leveraged strategy, the optimal Sharpe ratio portfolio is scaled to obtain a 200% exposure. The additional exposure is financed by borrowing at the risk-free rate.

1.3 Robust Mean-Variance Optimization

Classical mean-variance optimization is sensitive to estimation errors in expected returns. To address this issue, a robust formulation is used. The robust optimization problem can be written as

$$\min_w w^T Q w$$

subject to

$$w^T \mu \geq r_{target}, \quad \sum_{i=1}^n w_i = 1, \quad w_i \geq 0.$$

The target return and uncertainty parameters are selected to reflect a more conservative investment strategy that is less sensitive to estimation errors.

1.4 Benchmark Tracking Portfolio

The benchmark tracking strategy aims to replicate the performance of the S&P30 benchmark portfolio. Let w_b denote the vector of benchmark weights. The tracking error can be written as

$$(w - w_b)^T Q (w - w_b).$$

The optimization problem minimizes the tracking error:

$$\min_w (w - w_b)^T Q (w - w_b)$$

subject to

$$\sum_{i=1}^n w_i = 1, \quad w_i \geq 0.$$

In addition, a cardinality constraint is imposed so that the portfolio contains at most ten assets. This constraint reflects practical limitations on the number of positions held in the portfolio.

2 Performance Analysis for year 2024 to 2025

In this section we analyze the performance of the nine portfolio strategies over the 2024-2025 investment horizon. The strategies include the five mean-variance based strategies implemented in Assignment 1 and four additional strategies introduced in this assignment: equal risk contributions, leveraged maximum Sharpe ratio, robust mean-variance optimization, and benchmark tracking. The comparison is based on daily portfolio value, maximum drawdown, and the evolution of portfolio weights.

2.1 Daily Portfolio Value

The daily portfolio values for all nine strategies are shown in Figure 1 in the Appendix. All strategies start with approximately one million dollars, which corresponds to the value of the initial portfolio provided in the assignment. Several differences between strategies can be observed from the figure. The maximum expected return strategy achieves the highest portfolio value during the investment horizon. However, this strategy also exhibits very large fluctuations. The portfolio value increases rapidly during favourable market movements but also experiences noticeable drops. This behaviour reflects the fact that the strategy focuses purely on maximizing expected return without explicitly controlling portfolio risk.

In contrast, the minimum variance and equal risk contributions strategies produce much smoother portfolio value paths. The growth of these portfolios is more gradual and stable. Since these strategies aim to control portfolio risk, the resulting allocations are more diversified and less sensitive to individual asset movements.

The benchmark tracking strategy shows relatively steady growth during the period. Because the objective of this strategy is to replicate the S&P30 benchmark portfolio rather than aggressively optimize returns, its performance follows the general market trend more closely. The robust mean-variance optimization strategy demonstrates moderate but stable growth. Compared with classical mean-variance strategies, the robust approach incorporates uncertainty in the parameter estimates, which typically results in more conservative portfolio allocations.

2.2 Maximum Drawdown

To better evaluate downside risk, we examine the maximum drawdown for each strategy within each holding period. The results are shown in Figure 2 in the Appendix. The maximum expected return strategy experiences the largest drawdowns among all strategies. In some periods the drawdown exceeds 25%, indicating that the strategy can incur significant short-term losses despite achieving strong overall returns.

Other strategies show considerably smaller drawdowns. The minimum variance, equal risk contributions, and robust optimization portfolios generally remain within a narrower drawdown range. This suggests that these strategies provide better downside protection by distributing portfolio risk more evenly across assets.

The benchmark tracking strategy also exhibits moderate drawdowns. Since the portfolio composition closely follows the benchmark index, its risk profile is similar to the overall market behavior.

2.3 Dynamic Portfolio Allocation under Robust Optimization

Figure 3 in the Appendix illustrates the dynamic portfolio allocations produced by the robust mean-variance optimization strategy across the twelve rebalancing periods. The figure shows that portfolio weights change over time but remain relatively moderate. Most asset weights stay below approximately 20%, and only a few assets receive larger allocations in certain periods. This indicates that the robust optimization strategy avoids highly concentrated portfolios.

Another observation is that the weight changes between consecutive periods are relatively moderate. Compared with strategies such as maximum expected return or maximum Sharpe ratio, the robust strategy tends to produce more stable allocations. As a result, the amount of trading required between rebalancing periods is generally smaller. This behaviour is consistent with the objective of robust optimization, which aims to construct portfolios that are less sensitive to estimation errors in expected returns and covariance matrices.

Another important observation from Figure 3 is that the robust portfolio tends to change its asset weights more gradually over time. Compared with the minimum variance, maximum expected return, and maximum Sharpe ratio strategies, the adjustments between consecutive periods are typically smaller. This behaviour suggests that the robust optimization strategy may reduce the amount of trading required during portfolio rebalancing. Since the robust formulation explicitly accounts for uncertainty in the input parameters, it avoids extreme weight shifts that often occur in classical mean-variance optimization. As a result, the portfolio allocations remain relatively stable across periods.

From a practical perspective, reduced trading can be beneficial because it lowers transaction costs and leads to more stable portfolio management.

2.4 Comparison of Portfolio Strategies

In conclusion, the nine strategies exhibit different trade-offs between return and risk.

The maximum expected return portfolio generates the highest final portfolio value but also experiences the largest volatility and drawdowns. The minimum variance and equal risk contributions strategies provide more stable performance but achieve lower total returns.

The leveraged maximum Sharpe ratio strategy attempts to enhance returns by borrowing at the risk-free rate. While this can improve performance in favorable market conditions, it also increases exposure to market fluctuations.

The robust mean-variance optimization strategy offers a balanced compromise between risk and return. By accounting for uncertainty in parameter estimates, it produces portfolios that are more stable and less sensitive to estimation errors.

Finally, the benchmark tracking strategy provides consistent and diversified exposure to the market index. Although its return is not the highest among all strategies, it maintains relatively stable performance and moderate risk levels.

From a practical investment perspective, the robust mean-variance strategy appears to provide a good balance between return stability and risk control. Therefore, it would be a reasonable choice for managing a personal investment portfolio.

3 Performance during the 2008-2009 Financial Crisis

3.1 Daily Portfolio Value

The daily portfolio values of the nine strategies during the 2008-2009 period are shown in Figure 4. This time period corresponds to the global financial crisis, during which the market experienced significant volatility and large negative returns.

All strategies experience a substantial decline in portfolio value during late 2008 and early 2009, reflecting the severe market downturn. Among the strategies, the maximum expected return portfolio performs the worst. Because this strategy focuses heavily on assets with the highest estimated expected returns, it is highly sensitive to estimation errors and market shocks. As a result, the portfolio suffers large losses during the crisis and recovers more slowly afterwards.

In contrast, more diversified strategies such as the equal-weighted portfolio, equal risk contributions portfolio, and benchmark tracking portfolio demonstrate stronger resilience. These strategies maintain more balanced allocations across assets, which helps reduce exposure to extreme movements in individual stocks and allows the portfolios to recover more steadily after the market bottom in 2009.

3.2 Maximum Drawdown

To evaluate downside risk during the crisis period, we examine the maximum drawdown for each strategy across the twelve holding periods. The results are shown in Figure 5.

The maximum expected return strategy exhibits the largest drawdowns among all strategies. In several periods the drawdown exceeds 50%, indicating extremely high downside risk when market conditions deteriorate. Leveraged strategies also experience relatively large drawdowns because leverage amplifies both gains and losses.

In contrast, the minimum variance and equal risk contributions portfolios maintain smaller drawdowns throughout the crisis. These strategies explicitly control portfolio risk and distribute exposures more evenly across assets, which provides better downside protection during periods of severe market stress.

3.3 Dynamic Portfolio Allocation under Robust Optimization

Figure 6 illustrates the dynamic portfolio allocations generated by the robust mean-variance optimization strategy over the twelve rebalancing periods.

The portfolio weights change over time in response to updated estimates of expected returns and covariance matrices. However, the allocations remain relatively diversified and avoid extremely large positions in individual assets. This reflects the objective of robust optimization, which accounts for uncertainty in parameter estimation and penalizes overly concentrated allocations.

As a result, the robust portfolio exhibits more stable allocation patterns compared with aggressive return-maximizing strategies, contributing to more controlled risk exposure during the crisis period.

4 Performance of Strategies during 2022

4.1 Daily Portfolio Value

The daily portfolio values of the nine strategies during 2022 are shown in Figure 7. Compared with the extreme market turmoil observed during the 2008-2009 financial crisis, the fluctuations in 2022 are less severe but still reflect a challenging market environment characterized by volatility and periods of declining equity prices.

Several differences in strategy performance can be observed. The minimum variance portfolio achieves the most stable growth throughout the year and maintains the highest portfolio value for most of the trading period. This reflects the defensive nature of the strategy, which allocates more weight to low-volatility assets and therefore provides better protection during uncertain market conditions.

The maximum expected return portfolio again exhibits the poorest performance among the strategies. Because this strategy concentrates heavily on assets with the highest estimated expected returns, the resulting portfolio becomes highly sensitive to estimation errors and adverse market movements. Consequently, it experiences larger losses and recovers more slowly than the other strategies.

Diversified strategies such as the equally weighted portfolio and the equal risk contributions portfolio show moderate performance and relatively stable portfolio value paths. These strategies distribute risk more evenly across assets, which helps to reduce the impact of large price movements in individual securities.

4.2 Maximum Drawdown

To further evaluate downside risk, we examine the maximum drawdown for each strategy across the six holding periods of 2022. The results are shown in Figure 8.

The maximum expected return strategy again exhibits the largest drawdowns among all strategies, reaching approximately 30% in some periods. This highlights the vulnerability of aggressive return-maximizing portfolios during periods of market decline.

In contrast, the minimum variance portfolio maintains the smallest drawdowns across most periods, confirming its effectiveness in controlling downside risk. Other diversified strategies, including the equal risk contributions and robust optimization portfolios, also exhibit moderate drawdowns that remain significantly smaller than those observed in the maximum expected return strategy.

When compared with the 2008-2009 crisis period, the drawdowns observed in 2022 are substantially smaller. During the financial crisis, several strategies experienced drawdowns exceeding 50%, whereas the drawdowns in 2022 generally remain within a much narrower range. This indicates that although 2022 was a difficult year for equity markets, the level of systemic stress was considerably lower than that experienced during the global financial crisis.

4.3 Dynamic Portfolio Allocation under Robust Optimization

Figure 9 illustrates the dynamic portfolio allocations generated by the robust mean-variance optimization strategy across the six rebalancing periods in 2022. The figure shows that portfolio weights change over time in response to updated estimates of expected returns and covariances. However, similar to the results observed in previous periods, the allocations remain relatively diversified and avoid extremely concentrated positions in individual assets.

This behaviour reflects the objective of robust optimization, which explicitly incorporates uncertainty in parameter estimates and penalizes overly aggressive allocations. As a result,

the robust portfolio maintains stable and well-diversified allocations throughout the year, contributing to controlled risk exposure and relatively stable portfolio performance.

5 Conclusion

This study evaluates nine portfolio allocation strategies across different market environments, including the 2008-2009 financial crisis and the volatile market conditions in 2022.

The results show that strategies emphasizing diversification and risk control, such as the minimum variance and equal risk contributions portfolios, generally achieve more stable performance and smaller drawdowns. In contrast, the maximum expected return strategy tends to experience larger losses during market downturns.

Overall, the drawdowns during the 2008-2009 financial crisis are substantially larger than those observed in 2022, reflecting the more severe systemic shock during the global financial crisis.

Appendix

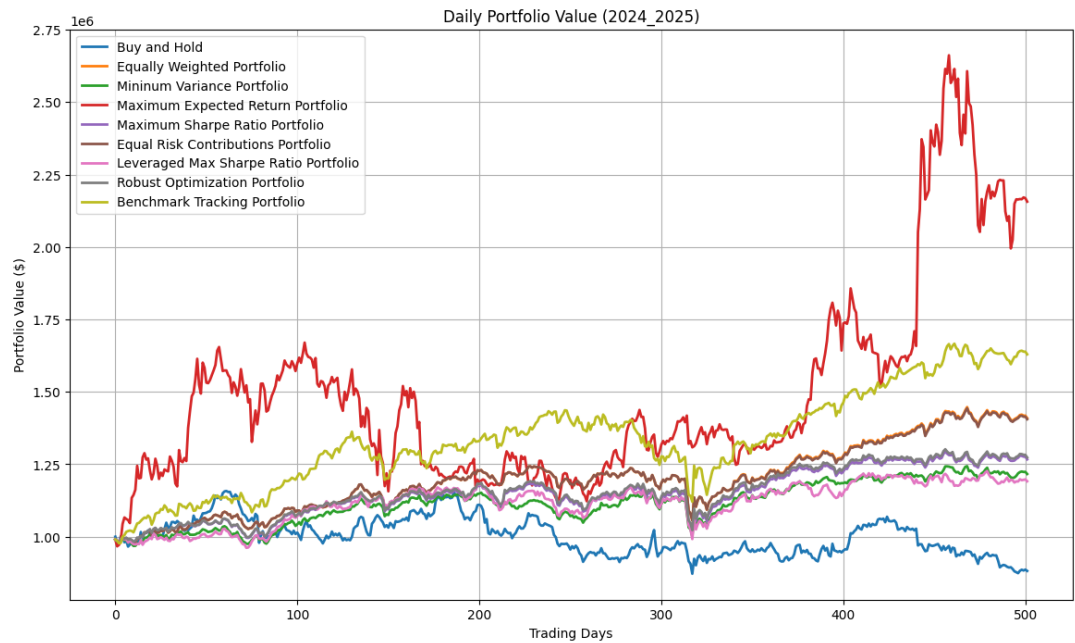


Figure 1: Daily Portfolio Value for the Nine Strategies (2024-2025)

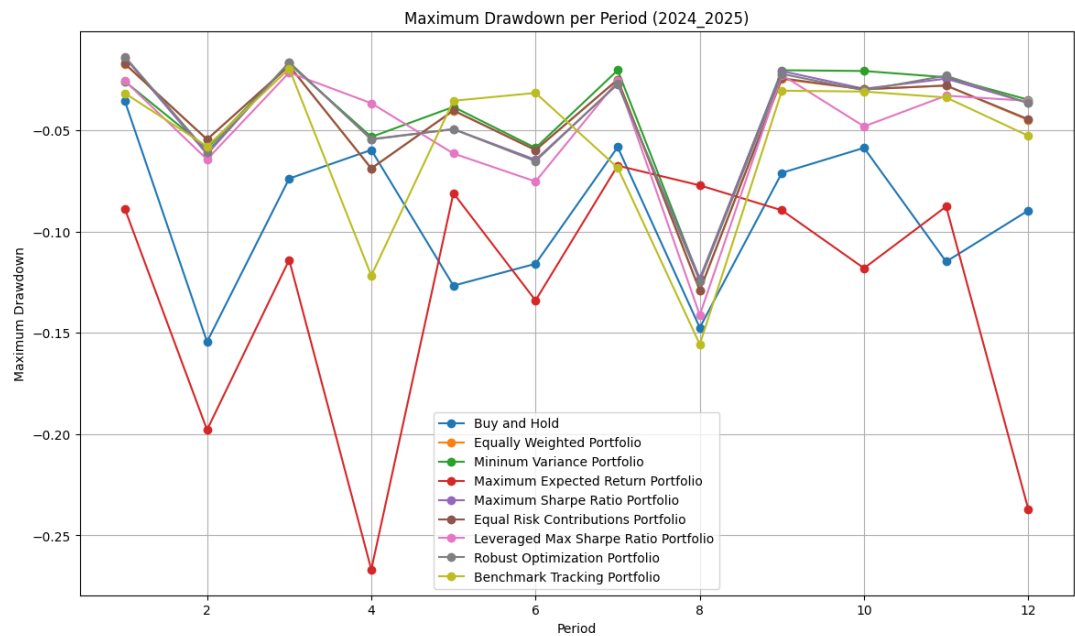


Figure 2: Maximum Drawdown per Period (2024-2025)

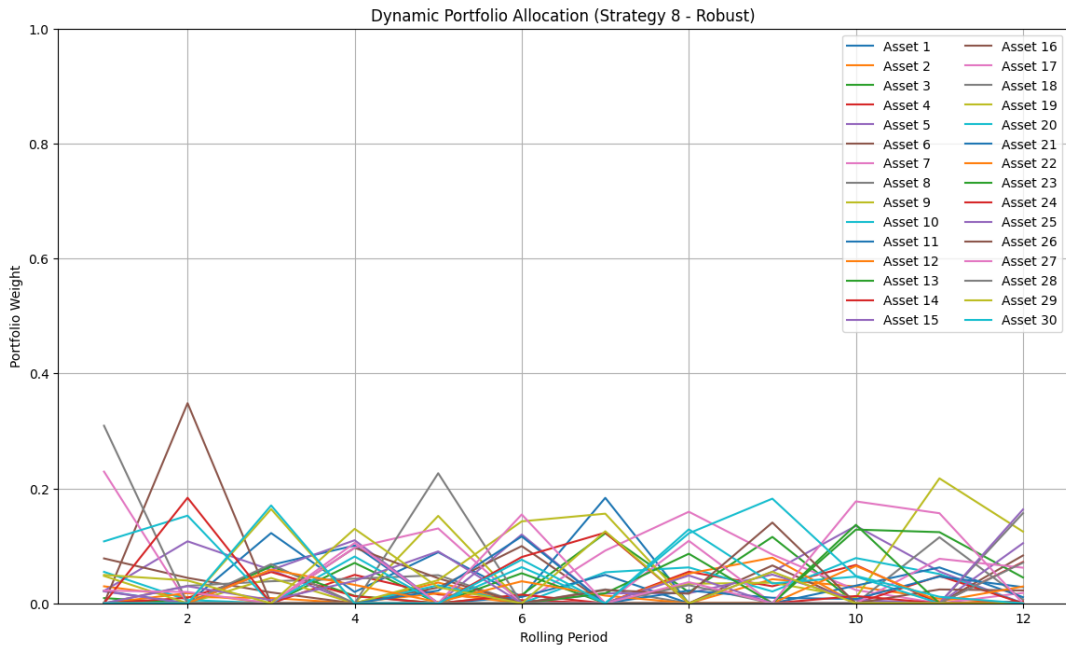


Figure 3: 2024-2025: Dynamic Portfolio Allocation (Strategy 8 - Robust)

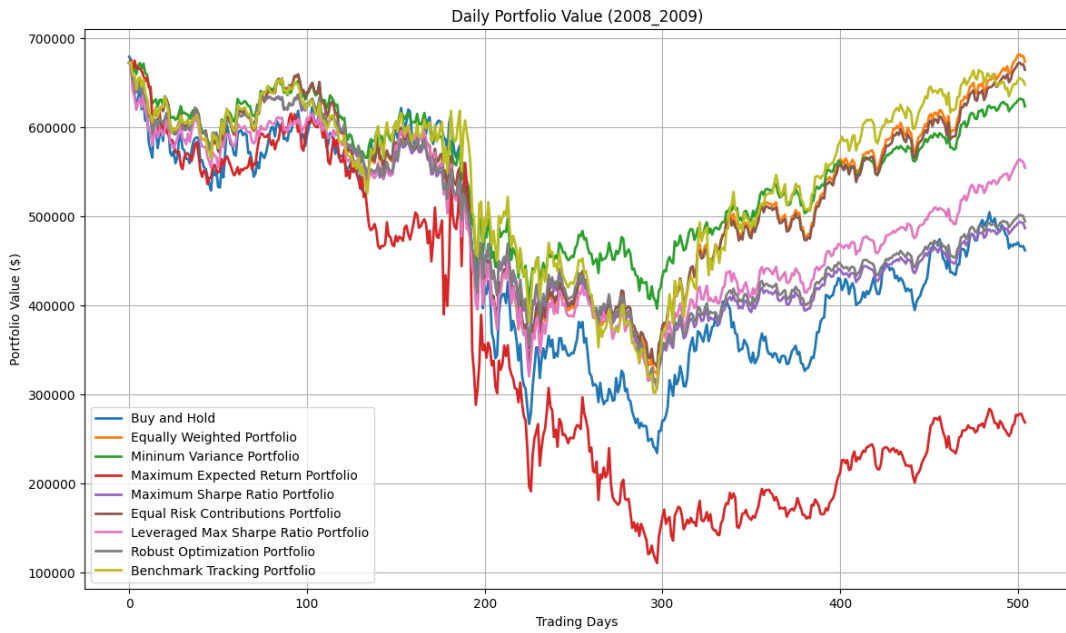


Figure 4: Daily Portfolio Value for the Nine Strategies (2008-2009)

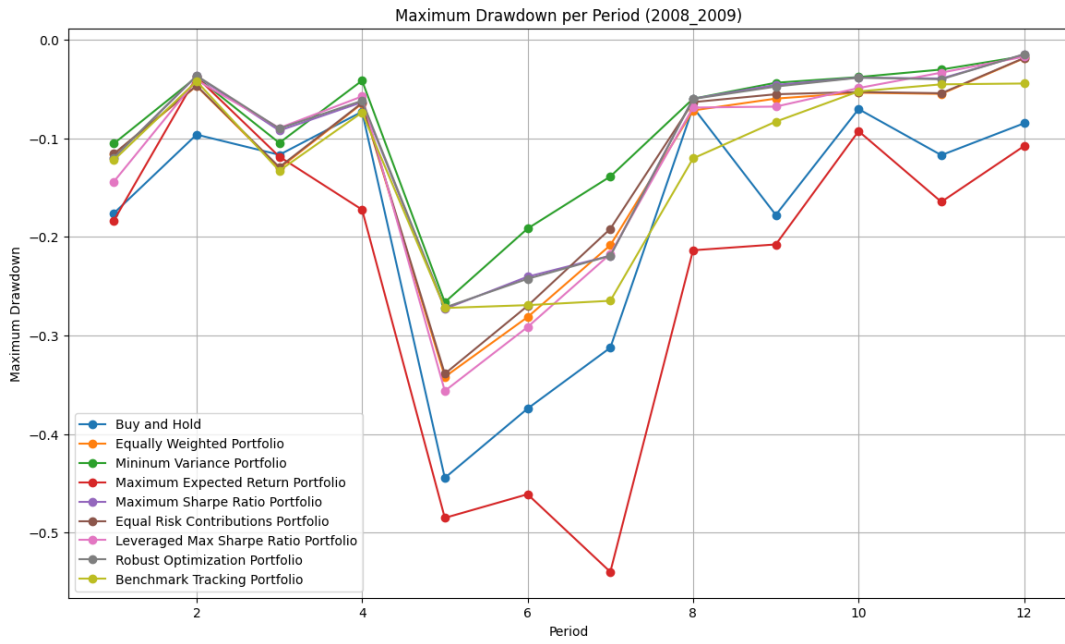


Figure 5: Maximum Drawdown per Period (2008-2009)

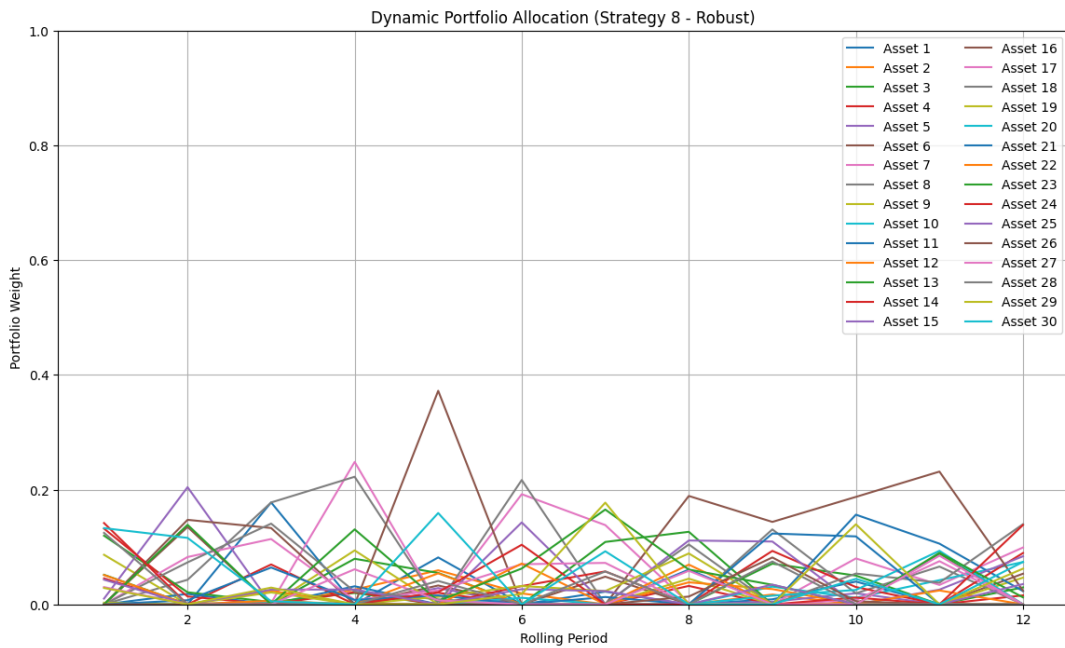


Figure 6: 2008-2009: Dynamic Portfolio Allocation (Strategy 8 - Robust)

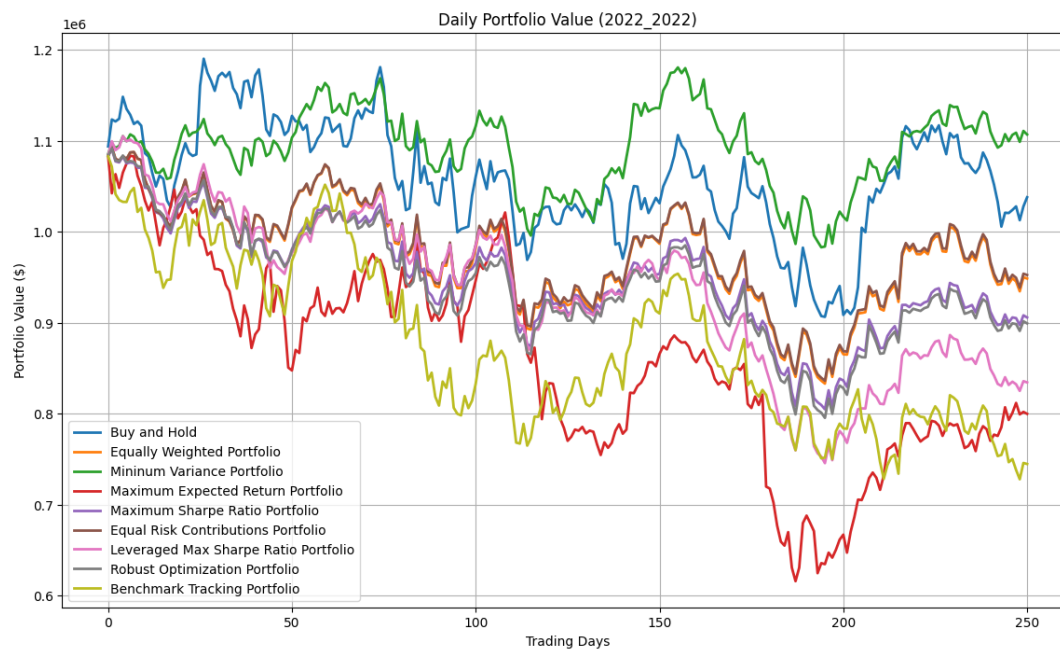


Figure 7: Daily Portfolio Value for the Nine Strategies (2022)

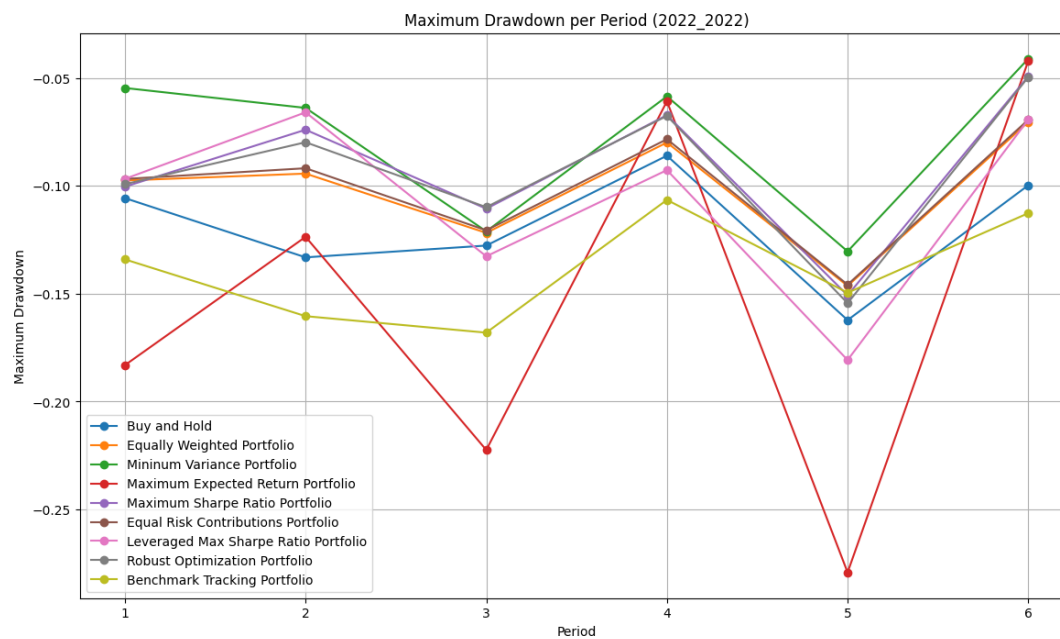


Figure 8: Maximum Drawdown per Period (2022)

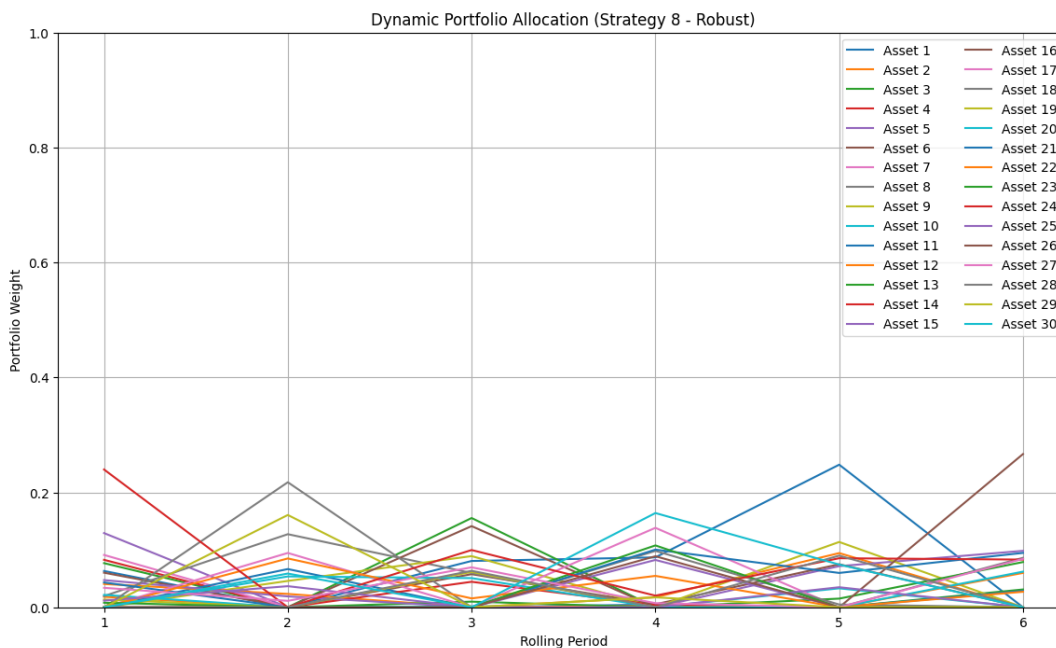


Figure 9: 2022: Dynamic Portfolio Allocation (Strategy 8 - Robust)

Appendix: Use of Generative AI

Generative AI tools were used in a limited and supportive manner during this assignment. ChatGPT was consulted primarily to confirm understanding of the assignment requirements, discuss possible report organization, and assist with debugging portions of Python code when intermediate results appeared inconsistent. It was also used to help format mathematical expressions into LaTeX.

Representative prompts included clarification of implementation logic, requests for debugging suggestions, and guidance on structuring the empirical analysis section.

All mathematical derivations were completed independently based on course material. All Python code was written and verified by the author, and final numerical results were checked through sanity tests and consistency checks (such as validating portfolio value evolution and transaction cost calculations).

After completing the implementation, OpenCode was used to confirm that the workflow satisfied the assignment specifications. Grammar corrections were performed using Microsoft Word's built-in tools.

AI tools were not used to generate empirical results or conduct the analysis itself. All interpretations and conclusions reflect the author's own verification and understanding.